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Aim of Experiment

To implement and compare Bubble Sort algorithms in C++ for sorting a list of data items.

Objectives

- To understand the concept of sorting techniques.
- To implement Bubble Sort algorithm using C++.
- To analyze the working principle and efficiency of these algorithms.
- To compare the time complexity of Bubble Sort algorithms.

Theory

Sorting

Sorting is the process of arranging data in a particular order (ascending or descending). Sorting makes searching and data processing more efficient.

Bubble Sort

Bubble Sort is a simple sorting algorithm that repeatedly compares adjacent elements and swaps them if they are in the wrong order.

Algorithm (Bubble Sort):

Step 1: Start
Step 2: Input array of n elements.
Step 3: Repeat for i = 0 to n-1
    For j = 0 to n-i-1
        If array[j] > array[j+1], swap them
Step 4: End

Time Complexity: $O(n^2)$
Best Case: O(n) (already sorted)
Worst Case: $O(n^2)$
Space Complexity: O(1)

Applications

Bubble Sort Applications:

- Simple and easy to implement for small datasets.
- Useful when simplicity is more important than efficiency.
- Helpful for teaching basic sorting concepts.

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Questions

Task-1

Manually **trace Bubble Sort** on a small list of integers.

i. Take a list of **05 unsorted integers** (distinct numbers).

ii. Apply the **Bubble Sort algorithm** to sort the list in **ascending order**.

References

1. Yedidyah Langsam, Moshe J Augenstein, Aaron M Tenenbaum – Data structures using C and C++ - PHI Publications (2nd Edition).
2. Ellis Horowitz, Sataraj Sahni- Fundamentals of Data Structures – Galgotia Books source.

Conclusion

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